

DRAFT

RESERVE FUND STUDY

**VILLAS AT WORTHINGTON, HOME OWNERS ASSOCIATION
N. ROYALTON, OHIO**

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1.0 INTRODUCTION

The Villas at Worthington, Home Owner Association authorized Criterium–Liszkay Engineers to conduct a Building Evaluation and Reserve Fund Study for the Villas at Worthington, located in N. Royalton, Ohio.

Studies of this nature are important to ensure that an Association has sufficient funds for long-term, periodic capital expenditure requirements. Anticipating large expenditures over an extended period of time through a structured analysis and scheduling process assists the Association in meeting financial requirements without increasing the service fees above permitted maximums, borrowing the funds or levying special financial assessments to the owners.

This report is designed to analyze components of the Association for which it is responsible and to assess a useful expected life and useful remaining life to those components. The anticipated scheduled repair or replacement of each component and the anticipated expense for the activity are then analyzed in conjunction with the current capital reserves funding program for the community.

This study should be considered as a full inspection since the entire site was inspected by an on-site inspection.

The present amount of funds in the reserve account is based on information from Renee Hambach, Associated Property Management, LLC and has not been audited by Criterium Engineers. Future project descriptions and funding amounts are based on information from the Association.

Typically, an association has two broad cash requirements: the general operating reserves and the capital repair and replacement reserves. In this report, we will focus on those items falling under the capital repair and replacement reserve criteria. We have projected a capital repair and replacement reserve for 20 years. The first 10 years are the most reliable. Such a study should be updated every five years.

This report is intended to be used as a tool to determine reserve fund allocation requirements for the community, to manage future Association obligations and to inform the Association of future financial needs in general. The report that follows has been prepared from the perspective of what an owner of this property would benefit from knowing. Some items, beyond those of immediate concern, may be discussed, because not including these issues could cause a distortion of the Association's situation. Therefore, the report should be read in its entirety in order to fully understand all of the information that has been obtained.

2.0 EXECUTIVE SUMMARY

The Villas at Worthington, Home Owners Association consists of grounds, walking paths, entry signs, ponds, fences and mailboxes.

The development was started in approximately 2008 with construction over the years and is complete. The houses are individually owned.

Based on our evaluation, the current level of funding of the reserve for this project is adequate. A more detailed analysis of the reserve fund has been provided in Appendix A.

There are several capital expenditures to be undertaken over the next 20 years. For your convenience, we have prepared the following summary of the condition of the major systems of the property.

PROPERTY SUMMARY			
SYSTEM	CONDITION	ACTIVITY REQUIRED	ANTICIPATED YEAR OF ACTIVITY
SITE			
Gravel Walking Paths	G	Repair	2020
Entry Signs	G	Repair	2031
Ponds	G	Dredge and repair bank	2051
Fences	G	Replace	2026
MECHANICAL AMENITIES			
Mail Box Stations	G-F	Replace	2051
OTHER			
Reserve Study	G	Update	2021

Table 2.1: Summary

3.0 PURPOSE & SCOPE

3.1 Purpose

The purpose of this study is to perform a reserve fund analysis and to determine a capital needs plan. It is intended to be used as a tool for the Villas at Worthington, Home Owners Association in determining the allocation requirements into the reserve fund in order to meet future anticipated capital expenditures for the Association to comply with Ohio Revised Code (ORC) 5312.

This report forecasts obligations for the Association 20 years into the future. This study is not intended to provide an inventory of all defects. It should be noted that events might occur that could have an effect on the underlying component or system useful life assumptions used in this study. Likewise, inevitable market fluctuations can have an impact on component or system replacement and repair costs. Therefore, a study such as this should be updated from time to time, usually on a five-year cycle, in order to reflect the most accurate needs and obligations of the community.

3.2 Scope

This study has been performed according to the scope as generally defined by the Villas at Worthington, Home Owner Association, Criterium-Liskay Engineers and the standards of the Community Associations Institute. The findings and recommendations are based on interviews with the community's management personnel and an investigation of the buildings and site.

The guidelines used to determine which physical components within the Master are to be included in the component inventory are based on the following general criteria:

1. The component must be a common element, or otherwise noted to be the responsibility of the Association to replace.
2. The funding for replacement should be from one source only, not funded from another area of the budget or through a maintenance contract.
3. The cost of replacement should be high enough to make it financially unsound to fund it from the operating budget.
4. Components, such as day to day painting, which are considered deferred maintenance, are most appropriately funded from the Operating Budget instead of Reserves.
5. Components with a useful life expectancy exceeding 30 years are not included.

Our reserve study analysis included evaluating the following association property:

Site and Grounds: The Villas at Worthington, Home Owners Association is approximately 40 acres that contain approximately 114 free standing family homes. The development was started in 2008 with construction over the years and is complete.

Private Streets, Sidewalks and Curbs: There is one entrance into the community off York Rd. in N. Royalton, Ohio. The concrete roads and curbs throughout the community are owned by the township. The sidewalks are maintained by the homeowners.

Building Common Elements: The individual homes are maintained by the homeowners.

Amenities: There is a post with wood fence at the entrance to the association. There is a gravel walking path with-in the community. There are four water detention ponds. There is a mailbox station.

This study estimates the funding levels required for maintaining the long term viability of the development. Our approach involves:

1. Examining association managed equipment, buildings and site facilities.
2. Predicting their remaining service life and approximating how frequently they will require repair or replacement for elements that have a predictable future life.
3. Estimating repair or replacement costs (in 2015 dollars) for each capital item and applying a 3% inflation rate.
4. Using data developed in Steps 1, 2 and 3 to project Capital Reserve balances for Years 1 through 20.

The statements in this report are opinions about the present condition of the subject community. We did not remove any surface materials, perform any destructive testing, or move any furnishings. This study is not an exhaustive technical evaluation. Such an evaluation would entail a significantly larger scope than this effort.

3.3 Sources of Information

Onsite inspections of the property occurred on the following dates:

- March 3, 2016

We based our cost estimates on some or all of the following:

- R.S. Means
- Our data files on similar projects
- Information from the Association
- Local contractors

For your reference, the following definitions may be helpful:

Excellent: Component or system is in "as new" condition, requiring no rehabilitation and should perform in accordance with expected performance.

Good: Component or system is sound and performing its function, although it may show signs of normal wear and tear. Some minor rehabilitation work may be required.

Fair: Component or system falls into one or more of the following categories: a) Evidence of previous repairs not in compliance with commonly accepted practice, b) Workmanship not in compliance with commonly accepted standards, c) Component or system is obsolete, d) Component or system approaching end of expected performance. Repair or replacement is required to prevent further deterioration or to prolong expected life.

Poor: Component or system has either failed or cannot be relied upon to

continue performing its original function as a result of having exceeded its expected performance, excessive deferred maintenance or state of disrepair. Present condition could contribute to or cause the deterioration of other adjoining elements or systems. Repair or replacement is required.

Adequate: A component or system is of a capacity that is defined as enough for what is required, sufficient, suitable and/or conforms to standard construction practices.

All ratings are determined by comparison to other buildings of similar age and construction type. Further, some details of workmanship and materials will be examined more closely in higher quality buildings where such details typically become more relevant.

All directions (left, right, rear, etc.), when used, are taken from the viewpoint of an observer standing in front of a building and facing it.

Repair/Replacement Reserves - Non-annual maintenance items that will require significant expenditure over the life of the buildings. Included are items that will reach the end of their estimated useful life during the course of this forecast, or, in the opinion of the investigator, will require attention during that time.

4.0 DESCRIPTION

The construction of Villas at Worthington, Home Owners Association started in approximately 2008. There is one entrance into the community, off York Rd. in N. Royalton, Ohio

At the entrance to the community there is a stone pier with signage and wood fences.

The concrete paved roads and concrete curbs throughout the community are owned by the township and not the responsibility of the association. The individual houses, yards, driveways and sidewalks are the responsibility of the homeowners.

There are gravel walking paths with wood fences with-in the community which are the responsibility of the association.

There are four detention ponds on site.

5.0 OBSERVATIONS

The following key observations were made about the current condition of the common elements of the property.

Site and Grounds

The stone signs at the entrance is in good condition. Funds for the eventual replacement of the stone sign is included in the study.

The wooden fences at the entrances and throughout the site are in good

condition. No funds for painting or repair of the fences are included in the study as this work should be in the maintenance portion of the operating budget, however funds for the eventual replacement of these fences are included in the study.

The gravel walking paths are in good condition. The gravel requires periodic repairs as the gravel settles. There are areas of walking paths in the woods which will require a higher maintenance costs due to the additional growth around the path. Funds for the periodic repairs are included in the study.

The ponds appeared to be in good condition. Over time, the ponds will fill with silt and dead plants and the banks will start to erode. Funds to dredge the ponds are included in the study. Repair to the banks is a maintenance item not a capital expenditure.

Amenities

The mail box stations are in good condition. The replacement of the mailboxes is included in the study.

6.0 RESERVE FUND ANALYSIS

Using software developed by Criterium Engineers and KPMG Peat Marwick, we have analyzed capital reserves draw-down for the projected capital expenditures to determine the amount needed. The following is a projected reserve fund analysis for non-annual items as discussed in the report. This projection takes into consideration a reasonable return on invested money and inflation. Please review this thoroughly and let us know of any changes that may be desired.

The intent of this reserve fund projection is to help the Association develop a reserve fund to provide for anticipated repair or replacements of various system components during the next 20 years.

The capital items listed are those that are typically the responsibility of the Home Owners Association however no formal list was provided but information was provided by Ohio Equities. Association by-laws vary, and therefore, which components is the responsibility of the owners and which are the responsibilities of the Association can vary. The Villas at Worthington, Home Owners Association should confirm that the items listed should be financed by the reserve fund. No taxes have been accounted for in this study.

This projection provides the following:

- An input sheet that defines all the criteria used for the financial alternatives, including the assumed inflation rate and rate of return on deposited reserve funds.
- A table that lists anticipated replacement and/or repair items complete with estimated remaining life expectancies, projected costs of replacement and/or repair, a frequency in years of when these items require replacement and/or repair and a projection based on

this frequency.

- A table and graph that represent end of year balances versus capital expenditures based on your current funding program and reserve balances, and alternatives to your current program. The provided graphs illustrate what effects the funding methods will have over the presented 20 year period versus the anticipated capital expenditures. Care should be taken in analyzing the graphs due to varying graphic scales that occur within each graph and between graphs.
- Note that based on our developed list of capital items and taking inflation into account the current funding is adequate.
- The Association should bear in mind that unanticipated expenditures can always arise and maintenance of a significant reserve fund balance can be viewed as a way to avoid special assessments.

Current Reserve Funding Rate: Based on the information received from Renee Hambach, Associated Property Management, \$20,656.00 is expected to be in the reserve fund at the beginning of 2016. Based on the calculation in this study this is adequate to be considered “fully funded.” See Alternative 1 to reduce the monthly fee and still be considered fully funded.

Current Funding Rate: \$540.00 monthly or \$6,480.00 annually.

- **Alternative 1:** Decrease the current contribution of \$540.00 per month to \$270.00 per month starting 2016. Maintain this over the next twenty years. This alternative will maintain a positive balance over the twenty year period.

Addendum A lists estimated capital reserves from 2016 through 2035.

7.0 CONCLUSION

The Villas at Worthington, Home Owners Association facilities are in good condition. With the ongoing existing maintenance programs for the communities’ largest assets, the current funding rate does appear to be adequate to fund the reserve fund over the next 20 years based on the assumptions about the line items contained in this report, including the 3% inflation rate.

We trust this answers any questions that may arise. If not, or if we can be of further assistance, please do not hesitate to call.

8.0 LIMITATIONS

The observations described in this study are valid on the dates of the investigation and have been made under the conditions noted in the report. We prepared this study for the exclusive use of The Villas at Worthington, Home Owners Association. Criterium–Liszkay Engineers does not intend any other individual or party to rely upon this study without our express written consent. If another individual or party relies on this study, they shall indemnify and hold Criterium–Liszkay Engineers harmless for any damages, losses or expenses they may incur as a result

of its use.

We did not remove surface materials, conduct any destructive or invasive testing, move furnishings or equipment or undertake any digging or excavation. Accordingly, we cannot comment on the condition of systems that we could not see, such as buried structures and utilities, nor are we responsible for conditions that could not be seen or were not within the scope of our services at the time of the investigation. We did not undertake to completely assess the stability of the buildings or the underlying foundation soil since this effort would require excavation and destructive testing. Likewise, this is not a seismic assessment.

We did not investigate the following areas:

- Buried utilities or infrastructure
- Concealed structural members or systems
- Any individual components
- None of the individual utilities

We do not render an opinion on uninvestigated portions of the community.

We did not perform any computations or other engineering analysis as part of this evaluation, nor did we conduct a comprehensive code compliance investigation. This study is not to be considered a warranty of condition, and no warranty is implied. The appendices are an integral part of this report and must be included in any review.

In our Reserve Fund Analysis, we have provided estimated costs. These costs are based on our general knowledge of building systems and the contracting and construction industry. When appropriate, we have relied on standard sources, such as Means Building Construction Cost Data, to develop estimates. However, for items that we have developed costs (e.g. structural repairs) no standard guide for developing such costs exists. Actual costs can vary significantly, based on the availability of qualified contractors to do the work, as well as many other variables. We cannot be responsible for the specific cost estimates provided.

We have performed no design work as part of this study, nor have we obtained competitive quotations or estimates from contractors since this also is beyond the scope of the project. The actual cost to remedy deficiencies and deferred maintenance items that we have identified may vary significantly from estimates and competitive quotations from contractors.

If you have any questions about this study or the reserve fund analysis, please feel free to contact us. Thank you for the opportunity to be of assistance to you.

Respectfully submitted,

Scott T. Malley, Architect

Appendix A: RESERVE FUND PROJECTIONS

Reserve Study Worksheet



General Information:

- 1 Organization: **Villas at Worthington, HOA**
 2 Address: **N. Royalton, OH**

3	Number of Units	1
4	Age of Building (in years)	5
5a	Study Period (in years)	20
5b	Normal Fiscal Year starts:	January 1, 2016
5c	Partial Fiscal Year starts:	January 1, 2016
5d	Partial Year Length:	12 months
6	Site Inspection Date	March 2, 2016
7	Reserve Funds at start	\$20,656
8	Rate of Return on invested Reserve Funds (%)	0.5%
9	Inflation Rate (%)	3.0%

10 Current Funding Levels

Existing Funding Levels

	Total/Month	Total Annual
Reserve Fund Contribution.....	\$540	\$6,480
	Years Out	Total Annual
Planned Special Assessment.....	0	\$0
Balance Computed.....	\$89,113	

Alternative 1 Reduced Funding

	Total/Month	Total Annual
Monthly Amount, (First Year).....	\$270	\$3,240
Monthly Amount, (Last Year).....	\$270	\$3,240
Balance Required Final Year.....	\$3,425	
Base Escalation %	0.00%	
	Years Out	Total/Year
Special Assessments:		
First Assessment.....	0	\$0
Second Assessment.....	0	\$0
Balance Computed.....	\$20,801	

Reserve Fund Worksheet

Fiscal Years:															
Normal: Jan 2016	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
Partial: Jan 2016 (12 months)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15

Existing Funding Levels

Beginning Reserve Fund Balance:	\$20,656	\$27,272	\$33,920	\$40,602	\$47,318	\$49,825	\$54,548	\$61,333	\$68,152	\$75,005	\$76,975	\$78,267	\$85,171	\$92,109	\$99,082
Revenue:	\$6,480	\$6,480	\$6,480	\$6,480	\$6,480	\$6,480	\$6,480	\$6,480	\$6,480	\$6,480	\$6,480	\$6,480	\$6,480	\$6,480	\$6,480
Special Assessments:	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
Investment Earnings:	\$136	\$169	\$202	\$235	\$248	\$271	\$305	\$339	\$373	\$383	\$389	\$424	\$458	\$493	\$499
Capital Expenditures:	\$0	\$0	\$0	\$0	\$4,221	\$2,029	\$0	\$0	\$0	\$4,893	\$5,577	\$0	\$0	\$0	\$5,672
Ending Reserve Balance:	\$27,272	\$33,920	\$40,602	\$47,318	\$49,825	\$54,548	\$61,333	\$68,152	\$75,005	\$76,975	\$78,267	\$85,171	\$92,109	\$99,082	\$100,389

Alternative 1, Level Funding with Steps

Average Cap. Expenditure \$3,425

Beginning Reserve Fund Balance:	\$20,656	\$20,759	\$20,863	\$20,967	\$21,072	\$16,936	\$14,982	\$15,057	\$15,132	\$15,207	\$10,366	\$4,813	\$4,837	\$4,861	\$4,885
Revenue:	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
Special Assessment #1:	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
Special Assessment #2:	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
Investment Earnings:	\$103	\$104	\$104	\$105	\$84	\$75	\$75	\$75	\$76	\$52	\$24	\$24	\$24	\$24	\$0
Capital Expenditures:	\$0	\$0	\$0	\$0	\$4,221	\$2,029	\$0	\$0	\$0	\$4,893	\$5,577	\$0	\$0	\$0	\$5,672
Ending Reserve Balance:	\$20,759	\$20,863	\$20,967	\$21,072	\$16,936	\$14,982	\$15,057	\$15,132	\$15,207	\$10,366	\$4,813	\$4,837	\$4,861	\$4,885	(\$787)

Alternative 2, Escalating Funding at 0% per Year

Beginning Reserve Fund Balance:	\$20,656	\$20,759	\$20,863	\$20,967	\$21,072	\$16,936	\$14,982	\$15,057	\$15,132	\$15,207	\$10,366	\$4,813	\$4,837	\$4,861	\$4,885
Revenue:	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
Special Assessment #1:	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
Special Assessment #2:	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
Investment Earnings:	\$103	\$104	\$104	\$105	\$84	\$75	\$75	\$75	\$76	\$52	\$24	\$24	\$24	\$24	\$0
Capital Expenditures:	\$0	\$0	\$0	\$0	\$4,221	\$2,029	\$0	\$0	\$0	\$4,893	\$5,577	\$0	\$0	\$0	\$5,672
Ending Reserve Balance:	\$20,759	\$20,863	\$20,967	\$21,072	\$16,936	\$14,982	\$15,057	\$15,132	\$15,207	\$10,366	\$4,813	\$4,837	\$4,861	\$4,885	(\$787)

Alternative 1, Reduced Funding

Beginning Reserve Fund Balance:	\$20,656	\$24,015	\$27,392	\$30,785	\$34,195	\$33,380	\$34,765	\$38,195	\$41,642	\$45,106	\$43,671	\$41,540	\$45,004	\$48,485	\$51,984
Revenue:	\$3,240	\$3,240	\$3,240	\$3,240	\$3,240	\$3,240	\$3,240	\$3,240	\$3,240	\$3,240	\$3,240	\$3,240	\$3,240	\$3,240	\$3,240
Special Assessment #1:	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
Special Assessment #2:	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
Investment Earnings:	\$119	\$136	\$153	\$170	\$166	\$173	\$190	\$207	\$224	\$217	\$207	\$224	\$241	\$259	\$248
Capital Expenditures:	\$0	\$0	\$0	\$0	\$4,221	\$2,029	\$0	\$0	\$0	\$4,893	\$5,577	\$0	\$0	\$0	\$5,672
Ending Reserve Balance:	\$24,015	\$27,392	\$30,785	\$34,195	\$33,380	\$34,765	\$38,195	\$41,642	\$45,106	\$43,671	\$41,540	\$45,004	\$48,485	\$51,984	\$49,799

Reserve Fund Worksheet

Fiscal Years:					
Normal: Jan 2016	2031	2032	2033	2034	2035
Partial: Jan 2016 (12 months)	16	17	18	19	20

Existing Funding Levels

Beginning Reserve Fund Balance:	\$100,389	\$89,006	\$95,964	\$102,956	\$109,983
Revenue:	\$6,480	\$6,480	\$6,480	\$6,480	\$6,480
Special Assessments:	\$0	\$0	\$0	\$0	\$0
Investment Earnings:	\$443	\$477	\$512	\$547	\$443
Capital Expenditures:	\$18,306	\$0	\$0	\$0	\$27,793
Ending Reserve Balance:	\$89,006	\$95,964	\$102,956	\$109,983	\$89,113

Alternative 1, Level Funding wi

Beginning Reserve Fund Balance:	(\$787)	(\$19,093)	(\$19,093)	(\$19,093)	(\$19,093)
Revenue:	\$0	\$0	\$0	\$0	\$0
Special Assessment #1:	\$0	\$0	\$0	\$0	\$0
Special Assessment #2:	\$0	\$0	\$0	\$0	\$0
Investment Earnings:	\$0	\$0	\$0	\$0	\$0
Capital Expenditures:	\$18,306	\$0	\$0	\$0	\$27,793
Ending Reserve Balance:	(\$19,093)	(\$19,093)	(\$19,093)	(\$19,093)	(\$46,886)

Alternative 2, Escalating Fundi

Beginning Reserve Fund Balance:	(\$787)	(\$19,093)	(\$19,093)	(\$19,093)	(\$19,093)
Revenue:	\$0	\$0	\$0	\$0	\$0
Special Assessment #1:	\$0	\$0	\$0	\$0	\$0
Special Assessment #2:	\$0	\$0	\$0	\$0	\$0
Investment Earnings:	\$0	\$0	\$0	\$0	\$0
Capital Expenditures:	\$18,306	\$0	\$0	\$0	\$27,793
Ending Reserve Balance:	(\$19,093)	(\$19,093)	(\$19,093)	(\$19,093)	(\$46,886)

Alternative 1, Reduced Funding

Beginning Reserve Fund Balance:	\$49,799	\$34,907	\$38,338	\$41,786	\$45,251
Revenue:	\$3,240	\$3,240	\$3,240	\$3,240	\$3,240
Special Assessment #1:	\$0	\$0	\$0	\$0	\$0
Special Assessment #2:	\$0	\$0	\$0	\$0	\$0
Investment Earnings:	\$174	\$191	\$208	\$225	\$103
Capital Expenditures:	\$18,306	\$0	\$0	\$0	\$27,793
Ending Reserve Balance:	\$34,907	\$38,338	\$41,786	\$45,251	\$20,801

Itemized Worksheet

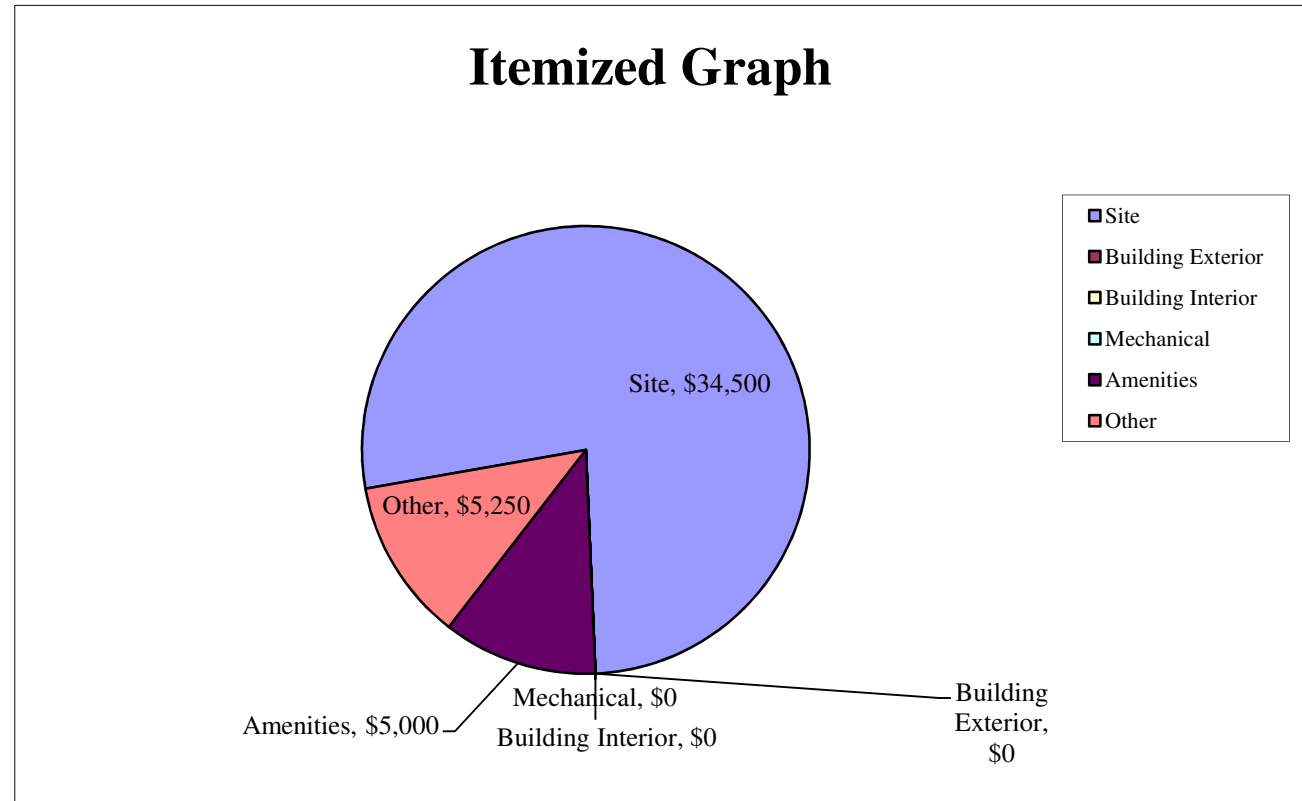
Capital Item To Be Replaced	Quantity	Unit cost	Reserve Requirement (*)	Beginning Balance	Frequency (yrs**)	Remaining Life (yrs)	Reserve Funding Monthly	Required Annual	Full Funding Balance	Information Source
Site										
Gravel Path - Repair	7,500	sf	\$0.50	\$3,750.00	\$1,825.45	5	4	\$40.09	\$481.14	\$750.00
Stone Entry Sign - Repair	2	ea	\$5,000.00	\$10,000.00	\$6,084.84	20	15	\$21.75	\$261.01	\$2,500.00
Wood Fences - Replace	100	lf	\$24.00	\$2,400.00	\$1,947.15	15	10	\$3.77	\$45.29	\$800.00
Pond - Dredge	71	cy	\$100.00	\$7,100.00	\$6,336.35	30	19	\$3.35	\$40.19	\$2,603.33
Building Exterior										
Building Interior										
Mechanical										
Amenities										
Mailbox Stations - Replace	1	ls	\$5,000.00	\$5,000.00	\$4,462.22	30	19	\$2.36	\$28.30	\$1,833.33
Other										
Reserve Study - Update	1	ls	\$1,750.00	\$1,750.00	\$0.00	5	5	\$29.17	\$350.00	\$0.00
Totals			\$30,000.00	\$20,656.00			\$100.49	\$1,205.93	\$8,486.67	
Total Over Term			\$44,750.00							
Total Over Term With Inflation			\$68,490.94							

* Costs are typically 10%±

** Reserve study is based on a 20 year projection of non-annual maintenance

Itemized Graph

Categories	Totals
Site	\$34,500
Building Exterior	\$0
Building Interior	\$0
Mechanical	\$0
Amenities	\$5,000
Other	\$5,250
Total	\$44,750



Itemized Funding

Categories	Reserve Requirement	Beginning Balance	Balance Requiring Funding	Monthly Reserve Funding Required	Annual Reserve Funding Required	Full Funding Balance	Simple Percent Funded
Site	\$34,500	\$16,194	\$18,306	\$69	\$828	\$6,653	
Building Exterior	\$0	\$0	\$0	\$0	\$0	\$0	
Building Interior	\$0	\$0	\$0	\$0	\$0	\$0	
Mechanical	\$0	\$0	\$0	\$0	\$0	\$0	
Amenities	\$5,000	\$4,462	\$538	\$2	\$28	\$1,833	
Other	\$5,250	\$0	\$5,250	\$29	\$350	\$0	
Totals	\$44,750	\$20,656	\$24,094	\$100	\$1,206	\$8,487	230.11 %

Annual Expense By Year

	Year: Year Number:	2016 1	2017 2	2018 3	2019 4	2020 5	2021 6	2022 7	2023 8	2024 9	2025 10	2026 11	2027 12	2028 13	2029 14	2030 15	2031 16
Site																	
Gravel Path - Repair		0	0	0	0	3,750	0	0	0	0	3,750	0	0	0	0	3,750	0
Stone Entry Sign - Repair		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	10,000
Wood Fences - Replace		0	0	0	0	0	0	0	0	0	0	2,400	0	0	0	0	0
Pond - Dredge		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Building Exterior																	
Building Interior																	
Mechanical																	
Amenities																	
Mailbox Stations - Replace		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Other																	
Reserve Study - Update		0	0	0	0	0	1,750	0	0	0	0	1,750	0	0	0	0	1,750
Total Costs		0	0	0	0	3,750	1,750	0	0	0	3,750	4,150	0	0	0	3,750	11,750
Total Costs Adjusted For 3% Inflation		0	0	0	0	4,221	2,029	0	0	0	4,893	5,577	0	0	0	5,672	18,306

Annual Expense By Year

	Year: Year Number:	2032 17	2033 18	2034 19	2035 20	
Site						
Gravel Path - Repair		0	0	0	3,750	
Stone Entry Sign - Repair		0	0	0	0	
Wood Fences - Replace		0	0	0	0	
Pond - Dredge		0	0	0	7,100	
Building Exterior						
Building Interior						
Mechanical						
Amenities						
Mailbox Stations - Replace		0	0	0	5,000	
Other						
Reserve Study - Update		0	0	0	0	
Total Costs		0	0	0	15,850	
Total Costs Adjusted For 3% Inflation		0	0	0	27,793	68490.94

Existing Funding Levels

Year	Year Number	Beginning Reserve Fund Balance	Fee Revenue	Special Assessments	Investment Earnings	Capital Expenditures	Ending Balance
2016	1	\$20,656	\$6,480	\$0	\$136	\$0	\$27,272
2017	2	\$27,272	\$6,480	\$0	\$169	\$0	\$33,920
2018	3	\$33,920	\$6,480	\$0	\$202	\$0	\$40,602
2019	4	\$40,602	\$6,480	\$0	\$235	\$0	\$47,318
2020	5	\$47,318	\$6,480	\$0	\$248	\$4,221	\$49,825
2021	6	\$49,825	\$6,480	\$0	\$271	\$2,029	\$54,548
2022	7	\$54,548	\$6,480	\$0	\$305	\$0	\$61,333
2023	8	\$61,333	\$6,480	\$0	\$339	\$0	\$68,152
2024	9	\$68,152	\$6,480	\$0	\$373	\$0	\$75,005
2025	10	\$75,005	\$6,480	\$0	\$383	\$4,893	\$76,975
2026	11	\$76,975	\$6,480	\$0	\$389	\$5,577	\$78,267
2027	12	\$78,267	\$6,480	\$0	\$424	\$0	\$85,171
2028	13	\$85,171	\$6,480	\$0	\$458	\$0	\$92,109
2029	14	\$92,109	\$6,480	\$0	\$493	\$0	\$99,082
2030	15	\$99,082	\$6,480	\$0	\$499	\$5,672	\$100,389
2031	16	\$100,389	\$6,480	\$0	\$443	\$18,306	\$89,006
2032	17	\$89,006	\$6,480	\$0	\$477	\$0	\$95,964
2033	18	\$95,964	\$6,480	\$0	\$512	\$0	\$102,956
2034	19	\$102,956	\$6,480	\$0	\$547	\$0	\$109,983
2035	20	\$109,983	\$6,480	\$0	\$443	\$27,793	\$89,113

Existing Funding Levels

Beginning Balance as of start of year beginning Jan 2016: \$20,656

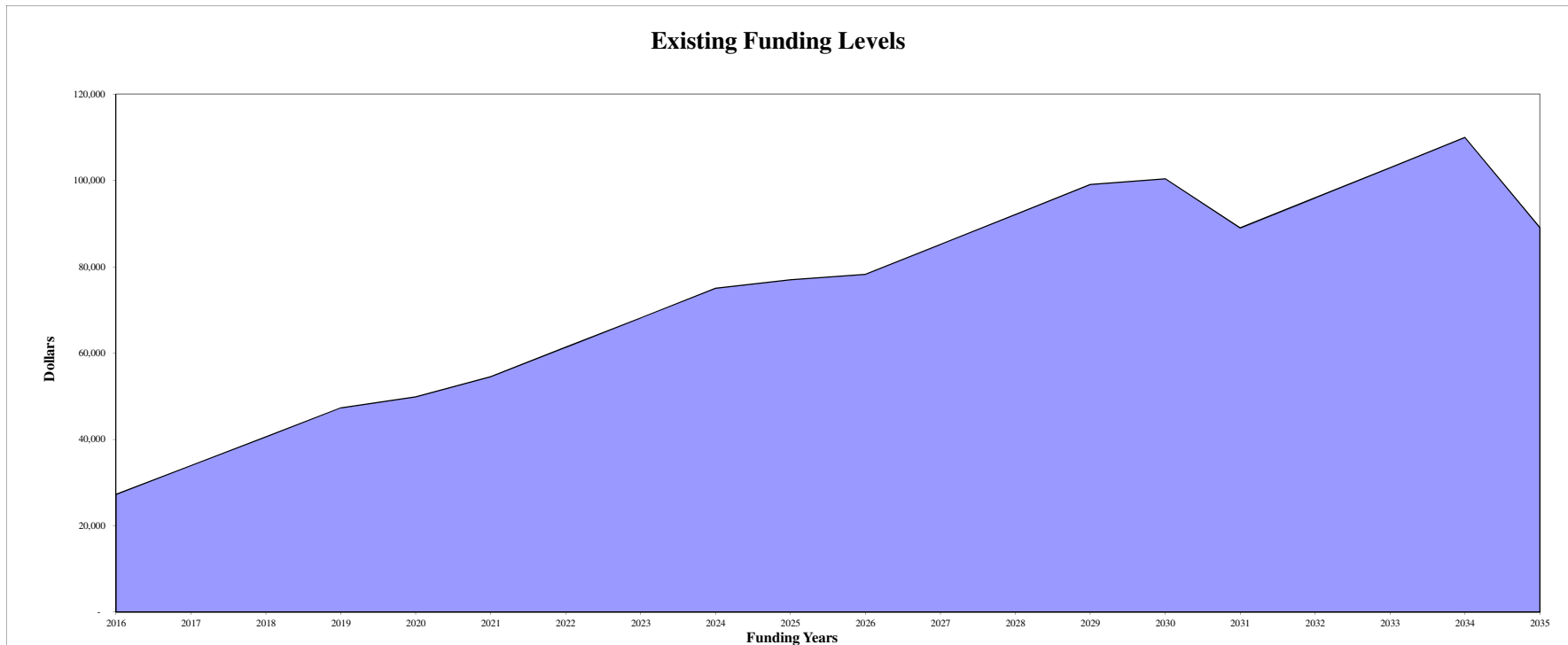
CONTRIBUTIONS	
AMOUNT	
\$6,480.00	per year
\$6,480.00	per unit per year
\$540.00	per month
\$540.00	per unit per month

SPECIAL ASSESSMENTS			
Totals			
Per Year	\$0	Per Unit	\$0

Projected Annual Funding and Expenditures:

Year:	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
Year Number:	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
End of Year Reserve Fund Balance	27,272	33,920	40,602	47,318	49,825	54,548	61,333	68,152	75,005	76,975	78,267	85,171	92,109	99,082	100,389
Capital Expenditures:	-	-	-	-	4,221	2,029	-	-	-	4,893	5,577	-	-	-	5,672
Total Revenue (all sources)	6,616	6,649	6,682	6,715	6,728	6,751	6,785	6,819	6,853	6,863	6,869	6,904	6,938	6,973	6,979

Year:	2031	2032	2033	2034	2035
Year Number:	16	17	18	19	20
End of Year Reserve Fund Balance	89,006	95,964	102,956	109,983	89,113
Capital Expenditures:	18,306	-	-	-	27,793
Total Revenue (all sources)	107,312	95,964	102,956	109,983	116,906



Alternative 1: Reduced Funding

Year	Year Number	Beginning Reserve Fund Balance	Fee Revenue	Special Assessments 1	Special Assessments 2	Investment Earnings	Capital Expenditures	Ending Balance
2016	1	\$20,656	\$3,240	\$0	\$0	\$119	\$0	\$24,015
2017	2	\$24,015	\$3,240	\$0	\$0	\$136	\$0	\$27,392
2018	3	\$27,392	\$3,240	\$0	\$0	\$153	\$0	\$30,785
2019	4	\$30,785	\$3,240	\$0	\$0	\$170	\$0	\$34,195
2020	5	\$34,195	\$3,240	\$0	\$0	\$166	\$4,221	\$33,380
2021	6	\$33,380	\$3,240	\$0	\$0	\$173	\$2,029	\$34,765
2022	7	\$34,765	\$3,240	\$0	\$0	\$190	\$0	\$38,195
2023	8	\$38,195	\$3,240	\$0	\$0	\$207	\$0	\$41,642
2024	9	\$41,642	\$3,240	\$0	\$0	\$224	\$0	\$45,106
2025	10	\$45,106	\$3,240	\$0	\$0	\$217	\$4,893	\$43,671
2026	11	\$43,671	\$3,240	\$0	\$0	\$207	\$5,577	\$41,540
2027	12	\$41,540	\$3,240	\$0	\$0	\$224	\$0	\$45,004
2028	13	\$45,004	\$3,240	\$0	\$0	\$241	\$0	\$48,485
2029	14	\$48,485	\$3,240	\$0	\$0	\$259	\$0	\$51,984
2030	15	\$51,984	\$3,240	\$0	\$0	\$248	\$5,672	\$49,799
2031	16	\$49,799	\$3,240	\$0	\$0	\$174	\$18,306	\$34,907
2032	17	\$34,907	\$3,240	\$0	\$0	\$191	\$0	\$38,338
2033	18	\$38,338	\$3,240	\$0	\$0	\$208	\$0	\$41,786
2034	19	\$41,786	\$3,240	\$0	\$0	\$225	\$0	\$45,251
2035	20	\$45,251	\$3,240	\$0	\$0	\$103	\$27,793	\$20,801

Alternative 1: Reduced Funding

Beginning Balance as of start of year beginning Jan 2016: \$20,656

CONTRIBUTIONS	
FIRST YR	LAST YR
\$3,240.00	\$3,240.00 per year
\$3,240.00	\$3,240.00 per unit per year
\$270.00	\$270.00 per month
\$270.00	\$270.00 per unit per month

SPECIAL ASSESSMENTS			
First Second	Per Year Per Year	Totals	
		\$0 \$0	Per Unit Per Unit
		\$0	\$0

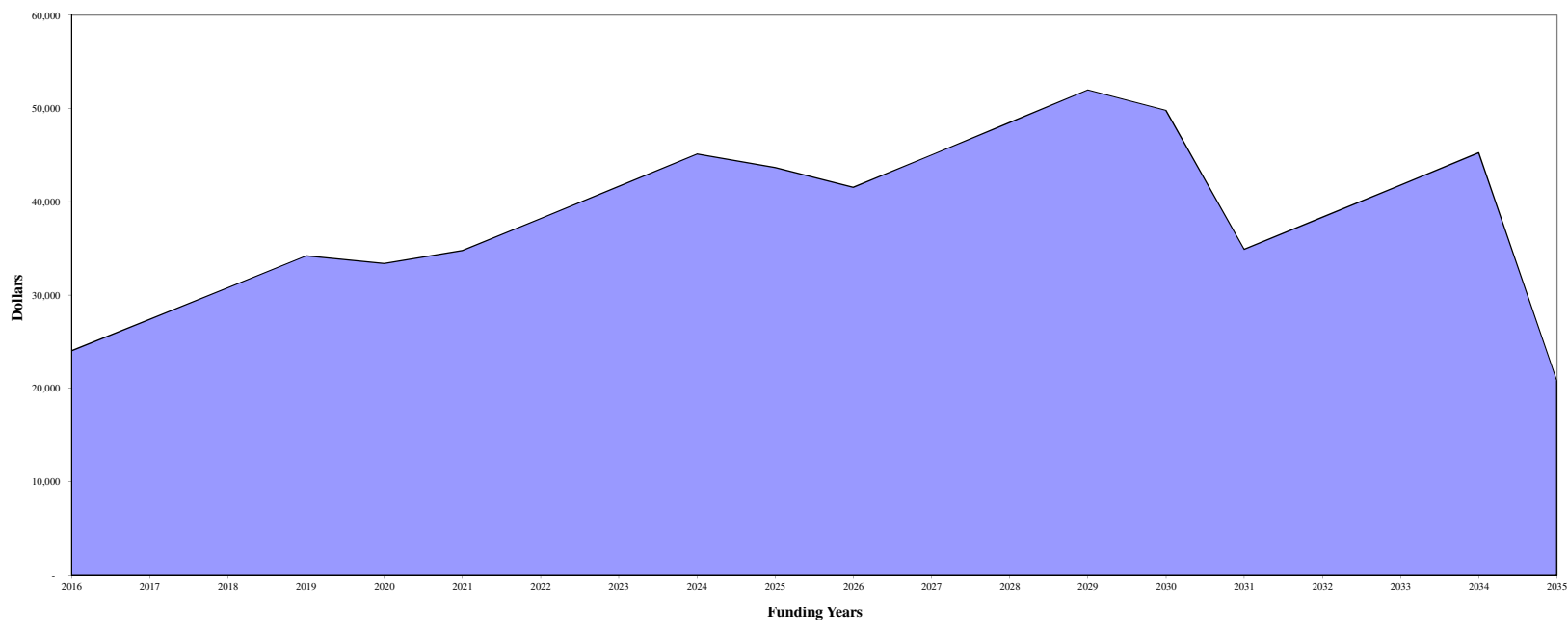
SETTINGS (analyzed by unit/month)		
Starting amount (\$):	270	
Increment by (%):	0	
Step (%):	0	
Every	3	year
Frequency:	3	time

Projected Annual Funding and Expenditures:

Year:	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
Year Number:	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
End of Year Reserve Fund Balance	24,015	27,392	30,785	34,195	33,380	34,765	38,195	41,642	45,106	43,671	41,540	45,004	48,485	51,984	49,799
Capital Expenditures:	-	-	-	-	4,221	2,029	-	-	-	4,893	5,577	-	-	-	5,672
Total Revenue (all sources)	3,359	3,376	3,393	3,410	3,406	3,413	3,430	3,447	3,464	3,457	3,447	3,464	3,481	3,499	3,488

Year:	2031	2032	2033	2034	2035
Year Number:	16	17	18	19	20
End of Year Reserve Fund Balance	34,907	38,338	41,786	45,251	20,801
Capital Expenditures:	18,306	-	-	-	27,793
Total Revenue (all sources)	3,414	3,431	3,448	3,465	3,343

Alternative 1: Reduced Funding

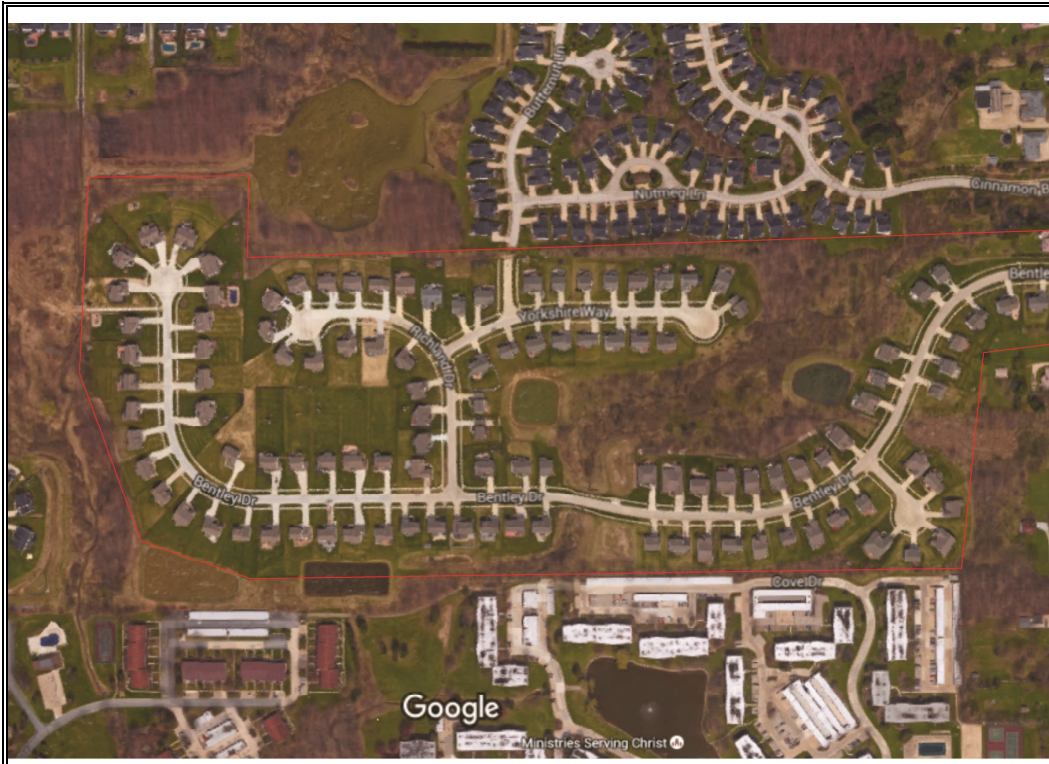


Appendix B: PROJECT PHOTOGRAPHS

Location:
Villas at Worthington
N. Royalton, OH

Photo Taken by:
Scott Malley, RA

Date:
March 2, 2016



Description:

Aerial view of the Villas at Worthington, Home Owners Association.

Photo Number

1



Description:

Entry sign and wood fence.

Photo Number

2

Location:
Villas at Worthington
N. Royalton, OH

Photo Taken by:
Scott Malley, RA

Date:
March 2, 2016



Description:
Sign for gravel walking path.

Photo Number
3



Description:
Gravel walking path and fences.

Photo Number
4

Location:
Villas at Worthington
N. Royalton, OH

Photo Taken by:
Scott Malley, RA

Date:
March 2, 2016



Description:
More gravel
walking path.

Photo Number
5



Description:
Wooden bridge.

Photo Number
6

Location:
Villas at Worthington
N. Royalton, OH

Photo Taken by:
Scott Malley, RA

Date:
March 2, 2016



Description:
Detention pond.

Photo Number
7



Description:
Mail box station.

Photo Number
8

Location:
Villas at Worthington
N. Royalton, OH

Photo Taken by:
Scott Malley, RA

Date:
March 2, 2016



Description:
Fence along
common area.

Photo Number
9

Appendix C: PROFESSIONAL QUALIFICATIONS

CRITERIUM-LISZKAY ENGINEERS

110 N. High St., Suite 207

Gahanna, Ohio 43230 614-418-7200, fax 614 418-7270 www.clengineers.com,

Scott T. Malley 614-282-7319 stmalley@hotmail.com

Education: The Ohio State University - Bachelors of Science in Architecture, 1983

Registered: Ohio #10445, 1993

Employment: **Self Employed** - January 2000 to Current

Working with Criterium-Liskay Engineers performing Condominium Reserve Studies, home inspections and commercial building analysis. Worked on retail stores, nail spas, restaurants, nightclubs and residential remodels. Took retail tenants from initial design through buildout, including site surveys, Architectural/MEP drawings and building permit processes. Consulted for Feinknopf Macioce Schappa Architects, Inc. performing assessments of various public school systems for the OFSC and as project manager for a series of medical office buildings. Consulted for Steiner + Associates as a Tenant Coordinator at The Greene Town Center, Beavercreek, Ohio, an 85 store lifestyle shopping center. Took clients from initial meetings, coordination of project documentation requirements and buildout through final closeouts. Used internal and web based programs to coordinate needs of multiple tenants with interoffice departments.

White Associates Architects, Inc. - January 1990 to December 1999

A small architecture firm working on a variety of projects including retail, commercial, residential, industrial and renovation. Participated in most phases of architecture including conceptual design, presentation drawings, code analysis, developing and producing construction documents and construction administration.